

KEVIN KIRUI

Healthcare AI & Workflow Systems Specialist

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Portfolio: app.notion.com/p/kevin-kirui-portfolio

PROFILE

Healthcare AI and workflow systems specialist with a biomedical engineering background and 2+ years spanning clinical engineering, healthcare data, AI evaluation, and software projects. Builds healthcare applications using Python, FastAPI, React, and SQL; evaluates AI-generated responses; validates healthcare data; and translates clinical operations problems into structured software and analytics solutions. Built 4 public healthcare applications, including a dashboard analyzing 10,483 health facilities across 47 Kenyan counties.

SELECTED WORK

Kenya Health Facilities Dashboard

Healthcare Analytics & Public Health Data

- Built and deployed using Python, FastAPI, React and health facility data, analyzing 10,483 facilities across all 47 Kenyan counties; developed facility distribution, service availability, and gap analysis.
- Improved mobile PageSpeed from 56 to 95 while maintaining 100 scores for Accessibility, Best Practices, and SEO.
- Reduced initial API requests from 6 to 1 with a combined dashboard-data endpoint; added API documentation and CSV export.

Clinical Workflow Signal Audit

Healthcare Workflow Intelligence

- Designed a synthetic ICU workflow audit using clinical signals and signal-to-action latency tracking to identify workflow risks and escalation delays.
- Built Monitor, Review, Escalate, and Activate tiers with role-based review; structured events into audit trails and SLA-based escalation routing.

Remote HealthAI Role Matcher

Research & Decision-Support System

- Built a repeatable workflow for identifying and prioritizing remote healthcare AI opportunities using scoring, filtering, and ranking logic.

PROFESSIONAL EXPERIENCE

Healthcare AI & Workflow Systems - Self-Directed Development & AI Evaluation

May 2024 - Present

- Build and deploy healthcare analytics, AI evaluation, and workflow intelligence applications using Python, FastAPI, React, and modern web tools.
- Evaluate AI-generated responses for accuracy, instruction-following, tone, usefulness, and healthcare safety across healthcare-related tasks.
- Translate clinical and operational problems into structured datasets, workflow logic, dashboards, and decision-support interfaces; document work through GitHub, READMEs, and API documentation.

HEALTHCARE EXPERIENCE

Kisii Teaching & Referral Hospital | Biomedical Engineering / Technical Support

Feb 2023 - Apr 2024

- Supported medical equipment maintenance, inspection, troubleshooting, and technical documentation across hospital departments.
- Worked alongside clinical teams to develop practical understanding of equipment availability, workflow dependencies, and downtime.

Kenyatta National Hospital & Tenwek Hospital | Biomedical Engineering Clinical Exposure

Gained clinical-environment exposure across ICU, renal, laboratory, diagnostic, and general hospital equipment workflows.

CORE SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Excel, Google Sheets, data cleaning, validation, CSV processing

AI & Workflow: LLM response evaluation, model output review, data annotation, instruction-following, healthcare safety, clinical workflow analysis

Software & Product: FastAPI, React, Tailwind CSS, GitHub, Vercel, API documentation, frontend/backend testing

Research: workflow documentation, structured research, technical reporting, Markdown, Notion

EDUCATION

Kenya Medical Training College, Nairobi | Diploma in Biomedical Engineering | Feb 2026

Kenya Medical Training College | Certificate in Medical Engineering | Dec 2023